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Surgical instruments and methods for performing tonsillectomy, adenoidectomy, and other surgical procedures

Abstract

A surgical instrument includes a housing, a shaft extending therefrom, an end effector assembly supported by the shaft, a movable handle, and a drive assembly. The drive assembly includes a translatable drive member for actuating the end effector assembly, and a torsion spring including first and second legs. The first leg is configured to translate through the housing in response to movement of the movable handle relative to the housing. The second leg is configured to translate through the housing in cooperation with the first leg to move the drive member longitudinally when a force acting on the drive member is less than a threshold force, and to remain in fixed position, thereby tensioning the torsion spring and retaining the drive member in fixed position when the force acting on the drive member is equal to or exceeds the threshold force.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application of U.S. patent application Ser. No. 17/065,033, filed on Oct. 7, 2020 and now U.S. Pat. No. 11,730,532, which is a continuation application of U.S. patent application Ser. No. 15/806,679, filed on Nov. 8, 2017 and now U.S. Pat. No. 10,828,084, which is a continuation application of U.S. patent application Ser. No. 14/719,422, filed on May 22, 2015 and now U.S. Pat. No. 9,918,779, the entire contents of each of which are hereby incorporated by reference herein.

BACKGROUND

Technical Field

(1) The present disclosure relates to surgical instruments and methods and, more particularly, to surgical instrument and methods for performing tonsillectomy, adenoidectomy, and other surgical procedures.

Background of Related Art

(2) The tonsils and adenoids are part of the lymphatic system and are generally located in the back of the throat. These parts of the lymphatic system are generally used for sampling bacteria and viruses entering the body and activating the immune system when warranted to produce antibodies to fight oncoming infections. More particularly, the tonsils and adenoids break down the bacteria or virus and send pieces of the bacteria or virus to the immune system to produce antibodies for fighting off infections.

(3) Inflammation of the tonsils and adenoids (e.g., tonsillitis) impedes the ability of the tonsils and adenoids to destroy the bacteria resulting in a bacterial infection. In many instances, the bacteria remain even after treatment and serve as a reservoir for repeated infections (e.g., tonsillitis or ear infections).

(4) A tonsillectomy and/or adenoidectomy may be performed when infections persist and antibiotic

treatments fail. Persistent infection typically leads to enlarged tonsil tissue which may need to be removed since in many cases the enlarged tissue causes airway obstruction leading to various sleep disorders such as snoring or, in some cases, sleep apnea. Some individuals are also born with larger tonsils that are more prone to cause obstruction. An adenoidectomy may also be required to remove adenoid tissue when ear pain persists, or when nose breathing or function of the Eustachian tube is impaired. Often times, tonsillectomy and adenoidectomy procedures are performed at the same time.

SUMMARY

(5) As used herein, the term “distal” refers to the portion that is being described which is further from a user, while the term “proximal” refers to the portion that is being described which is closer to a user. Further, to the extent consistent, any of the aspects described herein may be used in conjunction with any or all of the other aspects described herein.

(6) A surgical instrument provided in accordance with the present disclosure includes a housing, a shaft extending distally from the housing, an end effector assembly coupled at a distal end of the shaft, a movable handle coupled to the housing, and a drive assembly operably coupling the movable handle and the end effector assembly. The drive assembly includes a drive member and a torsion spring. The drive member is configured to translate through the shaft and relative to the end effector assembly to actuate the end effector assembly. The torsion spring includes a first leg and a second leg. The first leg is operably coupled to the movable handle and configured to translate longitudinally through the housing in response to movement of the movable handle relative to the housing. The second leg is operably coupled to the drive member and configured to translate longitudinally through the housing in cooperation with the first leg to thereby transfer longitudinal movement thereof into longitudinal movement of the drive member when a force acting on the drive member is less than a threshold force. The second leg is configured to remain in fixed position, thereby tensioning the torsion spring and retaining the drive member in fixed position, in response to longitudinal movement of the first leg when the force acting on the drive member is equal to or exceeds the threshold force.

(7) In an aspect of the present disclosure, the drive assembly further includes a slider including a housing portion operably retaining the torsion spring therein and a mandrel portion operably coupled to the movable handle. The slider is configured to translate longitudinally through the housing in response to movement of the movable handle relative to the housing.

(8) In another aspect of the present disclosure, the first leg of the torsion spring is engaged with the housing portion of the slider and the second leg of the torsion spring is movable relative to the housing portion of the slider. In such aspects, a body portion of the torsion spring may be rotatably supported on a post disposed within the housing portion of the slider. The post is configured to enable tensioning of the torsion spring in response to movement of the first leg relative to the second leg.

(9) In another aspect of the present disclosure, the first leg of the torsion spring is operably positioned relative to a block disposed within the housing to bias the drive member relative to the housing.

(10) In yet another aspect of the present disclosure, the movable handle is movable relative to the housing between an initial position, a compressed position, and an activated position. The second leg is configured to translate in cooperation with the first leg in response to movement of the movable handle between the initial and compressed positions. The second leg is configured to remain in fixed position in response to movement of the movable handle between the compressed and activated positions.

(11) In still another aspect of the present disclosure, the end effector assembly includes first and second jaw members, at least one of which is movable relative to the other between a spaced-apart position and an approximated position for grasping tissue therebetween. In such aspects, the drive member may be configured to translate through the shaft and relative to the end effector assembly

to move the first and second jaw members between the spaced-apart and approximated positions.

(12) In another aspect of the present disclosure, at least one of the first and second jaw members is adapted to connect to a source of energy for treating tissue grasped therebetween. In such aspects, an energy activation assembly disposed on the housing may also be provided. The energy activation assembly includes a switch configured to supply energy to the first and/or second jaw members. Further, the movable handle may be movable relative to the housing between an initial position, a compressed position, and an activated position. The second leg of the torsion spring is configured to remain in fixed position in response to movement of the movable handle between the compressed and activated positions and, in the activated position, at least a portion of the handle is configured to contact the energy activation assembly to activate the switch.

(13) In still another aspect of the present disclosure, the second leg is configured to translate in cooperation with the first leg in response to movement of the movable handle between the initial and compressed positions for applying an appropriate closure force to tissue grasped therebetween in the compressed position. Further, the appropriate closure pressure may be maintained, e.g., via tensioning of the torsion spring, during movement of the movable handle between the compressed and activated positions.

(14) In yet another aspect of the present disclosure, the torsion spring is pre-loaded to a less-tensioned state and wherein, in response to longitudinal movement of the first leg when the force acting on the drive member is equal to or exceeds the threshold force, the torsion spring is further tensioned to a more-tensioned state.

(15) Another surgical instrument provided in accordance with aspects of the present disclosure includes a movable handle, a drive member, a torsion spring, and a slider. The torsion spring includes a body, a first leg, and a second leg. The second leg of the torsion spring is engaged to the drive member. The slider includes a housing and a mandrel. The housing includes a post rotatably supporting the body of the torsion spring thereon. The housing is configured to permit movement of the second leg of the torsion spring relative to the housing. The mandrel extends from the housing and is operably coupled to the movable handle such that movement of the movable handle longitudinally translates the slider. When a force acting on the drive member is less than a threshold force, the slider and the second leg of the torsion spring are configured to translate longitudinally through the housing in cooperation with one another in response to movement of the movable handle, thereby transferring longitudinal movement of the slider into longitudinal movement of the drive member. When the force acting on the drive member is equal to or exceeds the threshold force, the torsion spring is tensioned in response to movement of the movable handle such that the second leg of the torsion spring and the drive member remain in fixed position during longitudinal movement of the slider.

(16) In an aspect of the present disclosure, the surgical instrument further includes an end effector assembly operably coupled to the drive member. In such aspects, the drive member is configured to translate relative to the end effector assembly to actuate the end effector assembly.

(17) In another aspect of the present disclosure, the end effector assembly includes first and second jaw members. At least one of the first and second jaw members is movable relative to the other between a spaced-apart position and an approximated position for grasping tissue therebetween in response to translation of the drive member relative to the end effector assembly.

(18) In still another aspect of the present disclosure, the force acting on the drive member corresponds to a closure pressure applied to tissue grasped between the first and second jaw members.

(19) In yet another aspect of the present disclosure, the movable handle is movable relative to the housing between an initial position, a compressed position, and an activated position. In such aspects, the slider and the second leg of the torsion spring are configured to translate longitudinally through the housing in cooperation with one another in response to movement of the movable handle between the initial and compressed positions. The second leg of the torsion spring and the

drive member remain in fixed position during longitudinal movement of the slider in response to movement of the movable handle between the compressed and activated positions.

(20) In still yet another aspect of the present disclosure, at least one of the first and second jaw members is adapted to connect to a source of energy for treating tissue grasped between the first and second jaw members. In such aspects, energy may be supplied to the at least one of the first and second jaw members in response to movement of the movable handle to the activated position.

(21) In an aspect of the present disclosure, the housing retains the first end of the torsion spring in fixed relation relative to the housing. Alternatively, the first leg of the torsion spring may be operably positioned to bias the drive member in a longitudinal direction.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Various aspects and features of the present disclosure described herein with reference to the drawings wherein:

(2) FIG. 1 is a perspective view of a surgical instrument provided in accordance with the present disclosure with jaw members of the end effector assembly of the surgical instrument disposed in a spaced-apart position;

(3) FIG. 2 is a top view of the surgical instrument of FIG. 1 with the jaw members disposed in an approximated position;

(4) FIG. 3 is a side view of the surgical instrument of FIG. 1 with the jaw members disposed in the approximated position;

(5) FIGS. 4-6 are side views of the surgical instrument of FIG. 1 illustrating various different configurations for operably grasping the surgical instrument;

(6) FIG. 7 is a rear, perspective view of the surgical instrument of FIG. 1 with the jaw members disposed in the approximated position and a portion of the housing removed to illustrate the internal components thereof;

(7) FIG. 8 is a front, perspective, partially-exploded view of the surgical instrument of FIG. 1 with the jaw members disposed in the approximated position and a portion of the housing removed to illustrate the internal components thereof;

(8) FIG. 9 is a rear, perspective view of the handle, trigger, and drive assemblies of the surgical instrument of FIG. 1 with a movable handle of the handle assembly disposed in an initial position and a trigger of the trigger assembly disposed in an un-actuated position;

(9) FIG. 9A is a rear, perspective view of a portion of the proximal end of the surgical instrument of FIG. 1 incorporating another configuration of a drive assembly provided in accordance with the present disclosure;

(10) FIG. 10 is a front, perspective view of the drive assembly, shaft, and end effector assembly of the surgical instrument of FIG. 1;

(11) FIG. 11 is an enlarged, perspective view of the area of detail indicated as “11” in FIG. 10;

(12) FIG. 11A is a perspective view of another slider assembly provided in accordance with the present disclosure and configured for use with the surgical instrument of FIG. 1;

(13) FIG. 12 is a rear, perspective, exploded view of the drive assembly, shaft, and end effector assembly of the surgical instrument of FIG. 1;

(14) FIG. 13 is a transverse, cross-sectional view taken along section line “13-13” of FIG. 3;

(15) FIG. 14 is an enlarged, perspective view of the distal end of the surgical instrument of FIG. 1;

(16) FIG. 15 is a side view of the distal end of the surgical instrument of FIG. 1 with the jaw members disposed in the spaced-apart position adjacent tissue to be grasped;

(17) FIG. 15A is a side view of the distal end of the surgical instrument of FIG. 1 with the jaw members disposed in the approximated position pressed against tissue to be spread and/or

dissected;

(18) FIG. 15B is a side view of the distal end of the surgical instrument of FIG. 1 incorporating another configuration of jaw members provided in accordance with the present disclosure and disposed in the approximated position pressed against tissue to be spread and/or dissected;

(19) FIG. 15B' is an enlarged, side view of the distal ends of the jaw members of FIG. 15B;

(20) FIG. 15C is a side view of the distal end of the surgical instrument of FIG. 1 incorporating yet another configuration of jaw members provided in accordance with the present disclosure and disposed in the approximated position pressed against tissue to be spread and/or dissected;

(21) FIG. 15C' is a top view of one of the jaw members of FIG. 15C;

(22) FIG. 15D is a side view of the distal end of the surgical instrument of FIG. 1 incorporating still another configuration of jaw members provided in accordance with the present disclosure and disposed in the approximated position pressed against tissue to be spread and/or dissected;

(23) FIG. 15D' is an enlarged, side view of the distal ends of the jaw members of FIG. 15D;

(24) FIG. 16 is a side view of the distal end of the surgical instrument of FIG. 1 with the jaw members disposed in the approximated position grasping tissue at the distal ends thereof;

(25) FIG. 16A is a side view of the distal end of the surgical instrument of FIG. 1 with the jaw members disposed in the spaced-apart position spreading and/or dissecting tissue;

(26) FIG. 16B is a side view of the jaw members of FIG. 15B disposed in the spaced-apart position spreading and/or dissecting tissue;

(27) FIG. 16C is a side view of the jaw members of FIG. 15C disposed in the spaced-apart position spreading and/or dissecting tissue;

(28) FIG. 16D is a side view of the jaw members of FIG. 15D disposed in the spaced-apart position spreading and/or dissecting tissue;

(29) FIG. 17 is an enlarged, perspective view of the area of detail indicated as "17" in FIG. 1;

(30) FIG. 18 is a front, perspective view of the distal end of the surgical instrument of FIG. 1 with the jaw members disposed in the spaced-apart position;

(31) FIG. 19 is a side, perspective view of the distal end of the surgical instrument of FIG. 1 with the jaw members disposed in the spaced-apart position;

(32) FIG. 20 is a side, perspective view of one of the jaw members of the surgical instrument of FIG. 1 with a portion thereof removed;

(33) FIG. 21 is a transverse, cross-sectional view of the jaw member of FIG. 20;

(34) FIG. 22 is a side, cross-sectional view taken along section line "22-22" of FIG. 1;

(35) FIG. 22A is a side, cross-sectional view of the proximal end of the surgical instrument of FIG. 1 incorporating another configuration of the torsion spring provided in accordance with the present disclosure;

(36) FIG. 23 is an enlarged, side, cross-sectional view of the area of detail indicated as "23" in FIG. 22;

(37) FIG. 24 is an enlarged, side, cross-sectional view of the area of detail indicated as "24" in FIG. 22;

(38) FIG. 25 is a perspective view of the surgical instrument of FIG. 1 with the movable handle disposed in a compressed position and, accordingly, the jaw members disposed in the approximated position;

(39) FIG. 26 is an enlarged, perspective view of the area of detail indicated as "26" in FIG. 25;

(40) FIG. 27 is a side, cross-sectional view of the proximal end of the surgical instrument of FIG. 1;

(41) FIG. 28 is an enlarged, side, cross-sectional view of the area of detail indicated as "28" in FIG. 27;

(42) FIG. 29 is a perspective view of the distal end of the drive and knife assemblies of the surgical instrument of FIG. 1 disposed in a position corresponding to the spaced-apart position of the jaw members;

- (43) FIG. **30** is a perspective view of the distal end of the drive and knife assemblies of the surgical instrument of FIG. **1** disposed in a position corresponding to the approximated position of the jaw members;
- (44) FIG. **31** is a perspective view of the distal end of the drive and knife assemblies as shown in FIG. **29** and further including one of the jaw members disposed in the spaced-apart position;
- (45) FIG. **32** is a perspective view of the distal end of the drive and knife assemblies as shown in FIG. **30** and further including one of the jaw members disposed in the approximated position;
- (46) FIG. **33** is a perspective view of the distal end of the drive and knife assemblies as shown in FIG. **29** and further including the jaw members disposed in the spaced-apart position;
- (47) FIG. **34** is a perspective view of the distal end of the drive and knife assemblies shown in FIG. **30** and further including the jaw members disposed in the approximated position;
- (48) FIG. **35** is a top, perspective view of the proximal end of the surgical instrument of FIG. **1** with the movable handle disposed in an activated position and a portion of the housing removed to illustrate the internal components thereof;
- (49) FIG. **36** is a top, perspective view of the proximal end of the surgical instrument of FIG. **1** with the trigger disposed in an actuated position and a portion of the housing removed to illustrate the internal components thereof;
- (50) FIG. **37** is a rear, perspective view of the handle, trigger, and drive assemblies of the surgical instrument of FIG. **1** with the movable handle disposed in the activated position and the trigger disposed in the actuated position;
- (51) FIG. **38** is a perspective view of the distal end of the drive and knife assemblies of the surgical instrument of FIG. **1** with the knife assembly disposed in an extended position;
- (52) FIG. **39** is a perspective view of the distal end of the drive and knife assemblies as shown in FIG. **38** and further including the jaw members disposed in the approximated position; and
- (53) FIG. **40** is a top, cross-sectional view of the distal end of the drive and knife assemblies and including the jaw members, as shown in FIG. **39**.

DETAILED DESCRIPTION

- (54) Referring generally to FIGS. **1-12**, a surgical instrument provided in accordance with the present disclosure is shown generally identified by reference numeral **10**. Instrument **10**, as described below, is configured for grasping, treating, and/or dissecting tissue and may find particular applicability for use in performing tonsillectomy and/or adenoidectomy procedures, although use of instrument **10** in various other surgical procedures is also contemplated and within the scope of the present disclosure.
- (55) With reference to FIGS. **1, 7, 8, and 12**, instrument **10** generally includes a housing **20**, a handle assembly **30**, a trigger assembly **70**, a shaft **80**, an end effector assembly **100**, a drive assembly **140**, a knife assembly **170**, and an energy activation assembly **190**. As detailed below, shaft **80** extends distally from housing **20** and supports end effector assembly **100** at distal end **85** thereof, drive assembly **140** operably couples handle assembly **30** with end effector assembly **100** to enable selective manipulation of jaw members **110, 120** of end effector assembly **100**, knife assembly **170** is operably coupled with trigger assembly **70** to enable selective translation of a knife blade **174** of knife assembly **170** relative to end effector assembly **100**, and energy activation assembly **190** enables energy to be selectively delivered to end effector assembly **100**.
- (56) Instrument **10** may also include an electrosurgical cable (not shown) that connects instrument **10** to a generator (not shown) or other suitable power source, although instrument **10** may alternatively be configured as a battery-powered instrument. The electrosurgical cable includes lead wires, e.g., lead wires **107** (FIG. **12**), extending therethrough that have sufficient length to extend through housing **20** and shaft **80** in order to operably couple the generator, energy activation assembly **190**, and end effector assembly **100** with one another to enable the selective supply of energy to electrically-conductive plates **112, 122** of jaw members **110, 120** of end effector assembly **100**, e.g., upon activation of activation switch **194** of energy activation assembly **190**.

(57) Referring to FIGS. 1-7, housing **20** houses the internal working components of instrument **10** and is formed from first and second housing components **22a**, **22b** configured to engage one another via a plurality of pin-aperture engagements **23** spaced around the perimeter of housing **20**. Housing **20** defines a pistol-style configuration having a longitudinally-extending barrel portion **24** and a fixed handle portion **26** that extends from barrel portion **24** in generally perpendicular orientation relative thereto.

(58) Housing **20**, movable handle **40** of handle assembly **30**, and trigger **72** of trigger assembly **70** are ergonomically configured to enable operable grasping of instrument **10** in a plurality of different positions. Housing **20**, more specifically, defines an elongated indentation **27** within barrel portion **24** on either side thereof, while a waist **28** is recessed annularly about handle portion **26** adjacent the interconnection between handle portion **26** and barrel portion **24**. Movable handle **40**, more specifically, defines a grasping portion **42** having an elongated proximal leg **43** that extends the length of fixed handle portion **26** of housing **20**, a proximal foot **44** disposed at the free end of proximal leg **43** and angled distally relative to proximal leg **43**, and an arcuate segment **45** disposed at the opposite end of proximal leg **43** and extending distally therefrom. Arcuate segment **45** culminates in a distal tail **46** and defines a sufficient diameter so as to operably receive a user's finger between distal tail **46** and proximal leg **43**. Trigger **72**, more specifically, includes a concave trigger surface **73** defining a saddle **74** configured to help retain a user's finger therein.

(59) With particular reference to FIGS. 4-6, and initially to FIG. 4, in a first operable grasping position, the user's hand is positioned such that the thumb is partially received within waist **28**, the tip of the index finger extends across trigger **72** and is partially received within saddle **74**, the middle finger extends across movable handle **40** and is positioned adjacent arcuate segment **45** of movable handle **40** between proximal leg **43** and distal tail **46**, and the ring finger and pinky are positioned distally of and adjacent to proximal leg **43**. In this position, waist **28** inhibits slipping of the thumb, saddle **74** inhibits slipping of the index finger, proximal leg **43** and distal tail **46** retain the middle finger therebetween to enable both proximal and distal movement of movable handle **40**, proximal leg **43** provides a surface against which the ring finger and pinky can be utilized to urge movable handle **40** proximally, and proximal foot **44** inhibits slipping of the ring finger and pinky off the free end of movable handle **40**.

(60) Referring to FIG. 5, in a second operable grasping position, the user's hand is positioned such that the index finger is partially received within elongated indentation **27** on the opposite side of housing **20**, the middle finger extends transversely across trigger **72** and is partially received within saddle **74**, the ring finger extends across movable handle **40** and is positioned adjacent arcuate segment **45** of movable handle **40** between proximal leg **43** and distal tail **46**, and the pinky is positioned distally of and adjacent to proximal leg **43**. In this position, elongated indentation **27** inhibits slipping of the index finger, saddle **74** inhibits slipping of the middle finger, proximal leg **43** and distal tail **46** retain the ring finger therebetween to enable both proximal and distal movement of movable handle **40**, and proximal leg **43** provides a surface against which the pinky can be utilized to urge movable handle **40** proximally.

(61) Referring to FIG. 6, in a third operable grasping position, the user's hand is positioned such that the thumb is wrapped around a free end of fixed handle portion **26** of housing **20**, the index finger and middle finger are positioned proximally of and adjacent the free end of proximal leg **43** of movable handle **40**, the ring finger extends across movable handle **40** and is positioned adjacent arcuate segment **45** of movable handle **40** between proximal leg **43** and distal tail **46**, and the pinky extends across trigger **72** and is partially received within saddle **74**. In this position, proximal foot **44** inhibits slipping of the index finger off the free end of movable handle **40**, proximal leg **43** and distal tail **46** retain the ring finger therebetween to enable both proximal and distal movement of movable handle **40**, proximal leg **43** provides a surface against which the index and middle fingers can be utilized to urge movable handle **40** proximally, and saddle **74** inhibits slipping of the pinky.

(62) With reference to FIG. 7, fixed handle portion **26** of housing **20** defines a bay **29a** configured

to receive and support energy activation assembly **190**, which is operable to initiate and terminate the delivery of energy to end effector assembly **100**. Energy activation assembly **190** includes a depressible button **192** that is mechanically coupled to a switch **194** mounted within bay **29a** of fixed handle portion **26** and is engagable by a button activation post **196** extending proximally from a proximal side of movable handle **40** upon movement of the movable handle **40** to the activated position, as detailed below. Switch **194** is configured to electrically communicate with end effector assembly **100** and the generator (not shown) via suitable electrical wiring, e.g., leads **107** (FIG. **12**), extending through housing **20**, shaft **80**, and/or an external cable (not shown) to enable energy to be supplied from the generator (not shown) to end effector assembly **100** upon activation of switch **194**.

(63) Referring additionally to FIG. **8**, barrel portion **24** of housing **20** defines a distal aperture **29b** (FIG. **7**) configured to receive proximal end **82** of shaft **80** therein, and an engagement feature (not shown) extending inwardly from each of first and second housing components **22a**, **22b** for receipt within opposed apertures **83** defined through proximal end **82** of shaft **80** for securing proximal end **82** of shaft **80** within barrel portion **24** of housing **20**. Shaft **80** extends distally from housing **20** and defines a generally rectangular cross-sectional configuration oriented such that the larger width dimension thereof extends laterally and the smaller height dimension thereof extends vertically. This configuration of shaft **80** relative to the orientation of jaw members **110**, **120** (FIG. **7**) provides an enhanced “line-of-sight” for visualizing the surgical site adjacent end effector assembly **100**. As described in greater detail below, shaft **80** includes a pair of spaced-apart clevis members **84** extending from the top and bottom walls, e.g., the larger width dimension walls, of shaft **80** at distal end **85** thereof, each of which defines an aperture **86** for receiving a pivot pin **103** to operably support end effector assembly **100** at distal end **85** of shaft **80**. In this configuration, apertures **86** are vertically-aligned with one another. Shaft **80** further includes, as noted above, opposed apertures **83** defined through the side walls, e.g., the smaller height dimension walls, of shaft **80** at proximal end **82** thereof for receiving engagement features (not shown) extending inwardly from first and second housing components **22a**, **22b** to secure proximal end **82** of shaft **80** within housing **20**.

(64) Barrel portion **24** of housing further includes a pair of pivot apertures **29c**, a longitudinal track **29d** (FIG. **13**), a pivot boss **29e**, and a retention pin **29f**. Each pivot aperture **29c** is defined on the inwardly-facing surface of one of first and second housing components **22a**, **22b** (only pivot aperture **29c** of first housing component **22a** is shown) and is configured to receive pivot pin **48** to pivotably couple movable handle **40** and trigger **72** to housing **20**. Longitudinal track **29d** (FIG. **13**) is defined on the inwardly-facing surface of first housing component **22a** and is configured to guide translation of drive assembly **140** relative to housing **20**. Pivot boss **29e** extends inwardly from first housing component **22a** and is configured to pivotably couple linkage **76** of trigger assembly **70** to housing **20**. Retention pin **29f** extends inwardly from first housing component **22a** and is configured to retain a fixed end **71a** of biasing member **71** of trigger assembly **70** in fixed position relative to housing **20**. The importance of these features of barrel portion **24** of housing **20** will become more apparent in view of the description below.

(65) Turning to FIGS. **7-13**, handle assembly **30** includes a movable handle **40** that is movable relative to fixed handle portion **26** of housing **20** between an initial position, a compressed position, and an activated position, as explained in greater detail below, to impart movement of jaw members **110**, **120** of end effector assembly **100** between a spaced-apart position and an approximated position for grasping tissue therebetween and for initiating the supply of energy to end effector assembly **100** for treating grasped tissue. Movable handle **40** includes grasping portion **42**, detailed above, which extends from housing **20** adjacent fixed handle portion **26**, and flange portion **47**, which extends upwardly into housing **20**. Flange portion **47** is pivotably coupled within housing **20** at the free end of flange portion **47** via pivot pin **48**. Pivot pin **48** is engaged within and extends between pivot apertures **29c** of first and second housing components **22a**, **22b** of housing **20** to

permit movable handle **40** to pivot about pivot pin **48** and relative to housing **20** between the initial position, the compressed position, and the activated position. Pivot pin **48** is disposed on one side of, e.g., above, drive assembly **140**, while grasping portion **42** of movable handle **40** is disposed on the other side of, e.g., below, drive assembly **140**, to provide a mechanical advantage when actuating movable handle **40**.

(66) Flange portion **47** of movable handle **40** further includes a cut-out **49** defined therein and an engagement bulge **51** protruding therefrom. Cut-out **49** is configured to slidably receive drive plate **142** of drive assembly **140** and knife plate **172** of knife assembly **170**. Engagement bulge **51** is configured to operably engage flange portion **47** of movable handle **40** with slider assembly **150** of drive assembly **140**, as detailed below.

(67) Drive assembly **140** includes drive plate **142** and slider assembly **150**. Drive plate **142** extends distally from housing **20** and through shaft **80** to operably engage end effector assembly **100** such that, as detailed below, translation of drive plate **142** through shaft **80** and relative to end effector assembly **100** pivots jaw members **110**, **120** of end effector assembly **100** between the spaced-apart and approximated positions. Slider assembly **150** operably couples flange portion **47** of movable handle **40** with drive plate **142** such that pivoting of movable handle **40** between the initial position and the compressed position pivots jaw members **110**, **120** of end effector assembly **100** between the spaced-apart and approximated positions, while ensuring application of an appropriate closure force or closure force within an appropriate closure force range to tissue grasped between jaw members **110**, **120** in the approximated position thereof.

(68) Slider assembly **150** includes a proximal housing **152**, a distal extension **154** extending distally from proximal housing **152**, and a mandrel **156** disposed at the distal end of distal extension **154**. Proximal housing **152** includes a post **153a** configured to receive a torsion spring **160** thereabout, a first slot **153b** configured to retain a first leg **161** of torsion spring **160** therein in fixed relation relative thereto, and a second slot **153c** configured to operably receive second leg **162** of torsion spring **160** therein. Proximal housing **152** further includes an abutment rib **153d** disposed thereon adjacent second slot **153c**, and a flange member **153e** configured for receipt within longitudinal track **29d** (FIG. **13**) of first housing component **22a** of housing **20**.

(69) Referring briefly to FIG. **11A**, another slider assembly **2150** provided in accordance with the present disclosure is similar to slider assembly **150** (FIGS. **10-11**) except for the configuration of the proximal housing and spring; thus, only these differences are detailed below. Proximal housing **2152** of slider assembly **2150** is configured to house a compression spring **2160** therein.

Compression spring **2160** defines a first end **2161** and a second end **2162**. First end **2161** of compression spring **2160** is engaged with a vertical plate. Second end **2162** of compression spring **2160** is engaged with an inner wall of proximal housing **2152**. In use, compression spring **2160** functions similar to torsion spring **160** (FIGS. **10-11**), as detailed below, except that, rather than being further tensioned via application of a torsional force thereto, compression spring **2160** is further tensioned via application of a compressive force thereto.

(70) Returning to FIGS. **7-13**, mandrel **156**, as noted above, is disposed at the distal end of distal extension **154** of slider assembly **150**. Mandrel **156** includes a pair of spaced-apart walls **157** defining a channel **158** therebetween. Channel **158** is configured to receive engagement bulge **51** of flange portion **47** of movable handle **40** while permitting vertical sliding of engagement bulge **51** within channel **158**. As a result of this configuration, upon pivoting of movable handle **40** between the initial, compressed, and activated positions, engagement bulge **51** is urged into contact with one of the walls **157** defining channel **158** to thereby translate slider assembly **150** within housing **20**. The vertical sliding of engagement bulge **51** within channel **158** during such urging ensures that slider assembly **150** is translated longitudinally within and relative to housing **20** despite the arcuate travel of movable handle **40** as movable handle is pivoted about pivot pin **48** relative to housing **20**.

(71) Drive plate **142** includes a flange **143** disposed at the proximal end thereof. Flange **143** defines

an aperture **144** configured to receive second leg **162** of torsion spring **160** therein such that translation of second leg **162** of torsion spring **160** relative to housing **20** effects corresponding translation of drive plate **142** relative to housing **20**. With respect to slider assembly **2150**, vertical plate **2163** is engaged within a slot **2144** defined within drive plate **142** (see FIG. **11A**), and functions in a similar manner as detailed below with respect to slider assembly **150**. Flange **143** further defines a proximal edge **145** configured to abut abutment rib **153d** of proximal housing **152** in a proximal-most position of drive plate **142** relative to slider assembly **150** to inhibit further proximal movement of drive plate **142** relative to slider assembly **150**.

(72) Drive plate **142**, as mentioned above, extends distally from housing **20** and through shaft **80** to operably engage end effector assembly **100**. Drive plate **142** is oriented similarly to shaft **80**, e.g., such that the width of drive plate **142** extends along the width dimension of shaft **80**. Drive plate **142** further defines a track edge **146** extending along a portion of each longitudinal side thereof. Track edges **146** are configured to slidably receive knife plate **172**, as detailed below. A cam-pin aperture **147** configured to receive a cam pin **105** associated with end effector assembly **100** is defined transversely through drive plate **142** towards the distal end of drive plate **142**. A mouth **149** configured to receive a pivot pin **103** associated with end effector assembly **100** is defined at the distal end of drive plate **142**.

(73) With momentary reference to FIG. **9A**, as an alternative to providing a slider assembly that operably retains a torsion spring therein for coupling to drive plate **142**, a torsion spring **1160** may be operably coupled between movable handle **40** and drive plate **142** without a slider assembly. More specifically, in some embodiments, torsion spring **1160** is mounted about a post **1153a** extending transversely from movable handle **40** and includes a first leg **1161** and a second leg **1162**. First leg **1161** of torsion spring **1160** is configured for receipt within an aperture **1144** defined through drive plate **142** to operably couple movable handle **40** with drive plate **142**, while second leg **1162** of torsion spring **1160** is fixed relative to movable handle **40** via abutment with a protrusion **1153c** thereof. In use, torsion spring **1160** operates similarly to torsion spring **160** (FIGS. **10-11**) and, thus, a separate description of the use of torsion spring **1160** is omitted as being superfluous.

(74) With reference to FIGS. **14-21**, as mentioned above, end effector assembly **100** is operably supported at distal end **85** of shaft **80** and includes opposing jaw members **110**, **120** pivotably coupled to one another and movable relative to one another and shaft **80** between a spaced-apart position and an approximated position for grasping tissue therebetween. Each jaw member **110**, **120** includes an electrically-conductive plate **112**, **122**, a jaw frame **113**, **123**, a spacer **115** (only spacer **115** of jaw member **110** is shown (FIG. **21**)), and an outer housing **118**, **128**, each of which is detailed below. Jaw members **110**, **120** define curved configurations, wherein jaw members **110**, **120** bend upwardly from a longitudinal axis of shaft **80**, e.g., towards the upper, larger width dimension wall of shaft **80**. This configuration facilitates use of instrument **10** in tonsillectomy and adenoidectomy procedures as well as other surgical procedures and allows for increased visualization of the surgical site in these and other procedures. Except where specifically noted otherwise, jaw members **110**, **120** define mirror-image configurations of one another.

(75) Jaw frames **113**, **123** of jaw members **110**, **120** each include a pair of spaced-apart proximal flanges **113a**, **123a** and a distal jaw support **113b**, **123b**. Jaw frames **113**, **123** are formed via stamping and made from stainless steel, although other manufacturing processes and/or materials for forming jaw frames **113**, **123** are also contemplated. Proximal flanges **113a** of jaw member **110** are spaced-apart further than proximal flanges **123a** of jaw member **120** so as to allow proximal flanges **123a** of jaw member **120** to be positioned between proximal flanges **113a** of jaw member **110** during assembly. Further, the proximal flanges **113a**, **123a** of each pair define aligned pivot apertures **114a**, **124a** and aligned cam slots **114b**, **124b**. Pivot pin **103** of end effector assembly **100** is configured for vertical insertion through apertures **86** of clevis members **84** of shaft **80** and pivot apertures **114a**, **124a** to pivotably couple jaw members **110**, **120** to shaft **80** and one another with

jaw members **110**, **120** being laterally movable, e.g., along the larger width dimension of shaft **80**, between the spaced-apart and approximated positions. Pivot pin **103** is configured to at least partially enter mouth **149** of drive plate **142** to permit drive plate **142** to slide further distally relative to end effector assembly **100** to a position wherein mouth **149** of drive plate **142** at least partially surrounds pivot pin **103**.

(76) The cam slots **114b** of proximal flanges **113a** of jaw member **110** are oppositely angled relative to the cam slots **124b** of proximal flanges **123a** of jaw member **120**. Cam pin **105** of end effector assembly **100** is configured for insertion through each cam slot **114b**, **124b** as well as cam-pin aperture **147** of drive plate **142** to operable couple drive plate **142** with jaw members **110**, **120** such that translation of drive plate **142** relative to jaw members **110**, **120** pivots jaw members **110**, **120** about pivot pin **103** and relative to one another and shaft **80** between the spaced-apart and approximated positions.

(77) With particular reference to FIGS. **19-21**, although only the features of jaw member **110** or jaw member **120** are described below and/or illustrated in the figures, it is noted that jaw members **110**, **120** defines a mirror-image configurations of one another (unless specifically contradicted herein) and, thus, any description and/or illustration of one jaw member **110**, **120** applies similarly to the other jaw member **110**, **120**.

(78) Distal jaw support **113b** of jaw frame **113** of jaw member **110** extends distally from proximal flange **113a** and defines a generally “L-shaped” configuration. Distal jaw support **113b** is configured to support electrically-conductive plate **112**, spacer **115**, and outer housing **118** of jaw member **110** thereon. However, distal jaw support **113b** do not extend distally the entire length of jaw member **110**. Rather, distal jaw support **113b** defines a length of about 50% to about 75% of the lengths of electrically-conductive plate **112**, spacer **115**, and outer housing **118** such that about 25% to about 50% of the lengths of these components extend distally beyond distal jaw support **113b**.

(79) Spacer **115** of jaw member **110** defines a generally “M-shaped” configuration, is formed from an electrically-insulative material, and is overmolded onto distal jaw support **113b** during a first overmold, although other manufacturing processes are also contemplated. Spacer **115** defines a body **116a** and a pair of wings **116b** surrounding body **116a**. Spacer **115** is positioned to electrically-isolate electrically-conductive plate **112** and distal jaw support **113b** from one another. A knife slot **116c** extends longitudinally through body **116a** of spaced **115** and is generally centered relative to body **116a**. Knife slot **116c** is open only to the top of spacer **115**, except for the distal portion thereof, which extends beyond distal jaw support **113b** and is open on both the top and bottom sides thereof to provide a window **116d**. A support-receiving channel **116e** extends longitudinally through body **116a** at a position laterally offset relative to knife slot **116c** so as to not interfere therewith. Support-receiving channel **116e** is open to the bottom of spacer **115** and is configured to receive the upright of the “L-shaped” distal jaw support **113b** upon the first overmolding of spacer **115** thereabout. Body **116a** of spacer **115** further defines a tunnel **116f** configured to permit passage of lead wire **107** therethrough.

(80) The electrically-conductive plate **112**, **122** of each jaw member **110**, **120** defines a generally planar tissue-contacting surface **112a**, **122a**, an elongated slot **112b**, **122b** extending through the respective tissue-contacting surface **112a**, **122a**, a pair of legs **112c**, **122c** extending downwardly from each side of the respective tissue-contacting surface **112a**, **122b**, and a distal edge **112d**, **122d** disposed at the distal end of the respective tissue-contacting surface **112a**, **122a**. Electrically-conductive plates **112**, **122** extend from the proximal heels of jaw members **110**, **120**, e.g., the interface between flanges **113a**, **123a** and the distal portions of jaw members **110**, **120**, to the distal tips of jaw members **110**, **120**. Jaw housing **118** of jaw member **110** includes a pair of proximal tissue stops that extend therefrom about either side of jaw member **120** such that, in conjunction with the positioning of electrically-conductive plates **112**, **122** at the proximal heel of jaw members **110**, **120**, grasping of tissue proximally of electrically-conductive plates **112**, **122** is inhibited.

(81) Tissue-contacting surfaces **112a**, **122a** define a plurality of spaced-apart recesses **112e**, **122e** therein that facilitate grasping tissue. Tissue-contacting surface **112a** of electrically-conductive plate **112** of jaw member **110** and/or tissue-contacting surface **122a** of electrically-conductive plate **122** of jaw member **120** may further include a plurality of stop members **122f** disposed thereon. Stop members **122f** may be constructed of a heat-resistant ceramic deposited onto the tissue-contacting surfaces **112a**, **122a**, an electrically non-conductive plastic molded onto tissue-contacting surfaces **112a**, **122a**, an electrical conductive material isolated from the respective tissue-contacting surface **112a**, **122a**, or may be formed from and/or manufactured in any other suitable fashion.

(82) Each wing **116b** of spacer **115** of jaw member **110** defines a slot **116g**, open at the top end thereof, that is configured for receiving one of the legs **112c** of electrically-conductive plate **112**. Wire **107**, which extends through tunnel **116f** defined within spacer **115** is configured to electrical connect to an underside of electrically-conductive plate **112** towards the distal end thereof for enabling the selective supply of energy thereto. Wire **107** is configured to extend proximally through shaft **80** and into housing **20**, ultimately coupling to energy activation assembly **190** (FIG. 7) and/or extending through the cable (not shown) to couple to the generator (not shown).

(83) Outer housings **118**, **128** are formed about jaw members **110**, **120** via a second overmold process, such that each outer housing **118**, **128** partially encloses respective jaw members **110**, **120** with the exception of a portion of the distal jaw support **113b**, **123b** thereof and the tissue-contacting surface **112a**, **122a** thereof, which remain exposed. Further, legs **112c**, **122c** of electrically-conductive plates **112**, **122** of jaw members **110**, **120** and the spacers **115** (only spacer **115** of jaw member **110** is shown) thereof each define a plurality of fill-apertures **122g** (only fill-apertures **122g** of electrically-conductive plate **122** of jaw member **120** are illustrated) that, upon overmolding of outer housings **118**, **128** about respective jaw members **110**, **120** are filled with the overmolded material forming outer housings **118**, **128** to lock the components of each jaw member **110**, **120** in an assembled condition. Further, outer housings **118**, **128** define lengths extending along the sides of respective jaw members **110**, **120** and thicknesses that decrease in the proximal-to-distal direction along the lengths thereof. Outer housings **118**, **128** also define windows **119**, **129** that align with and communicate with the windows **116d** of the respective spacers **115** (only spacer **115** of jaw member **110** is illustrated) and the knife slots **112b**, **122b** of the respective electrically-conductive plate **112**, **122** thereof so as to define an opening **131**, **132** extending through the distal portion of each jaw member **110**, **120** transversely relative to the plane defined by the respective tissue-contacting surface **112a**, **122a**.

(84) With outer housings **118**, **128** formed about jaw members **110**, **120**, respectively, distal edges **112d**, **122d** of electrically-conductive plates **112**, **122** overlap the distal ends of outer housings **118**, **128** such that, as illustrated in FIGS. 15 and 16, distal edges **112d**, **122d** can be utilized to pinch tissue therebetween. In particular, this configuration enables pinching of planar tissue structures that lack substantial protruding portions that would otherwise enable grasping, such as the tissue wall illustrated in FIGS. 15 and 16.

(85) Referring to FIGS. 15A and 16A, jaw members **110**, **120** may further be utilized to spread and/or dissect tissue. In order to do so, with jaw members **110**, **120** disposed in the approximated position, end effector assembly **100** may be manipulated such that the distal tips of jaw members **110**, **120** are pressed into contact with tissue to be spread and/or dissected, as shown in FIG. 15A. Thereafter, jaw members **110**, **120** are moved to the spaced-apart position such that the distal ends of outer housings **118**, **128** of jaw members **110**, **120**, respectively, push tissue in opposite directions, thus spreading and/or dissecting tissue. Various configurations of the distal ends of outer housings **118**, **128** of jaw members **110**, **120** to further facilitate spreading and/or dissecting tissue are detailed below. As also detailed below, drive assembly **140** (FIG. 12) defines a pre-loaded configuration wherein drive assembly **140** (FIG. 12) is always under tension, such that backlash upon moving jaw members **110**, **120** from the approximated position back to the spaced-apart

position is eliminated. Such a configuration facilitates spreading and/or dissecting tissue by allowing for a more smooth and consistent transition of jaw members **110**, **120** from the approximated position back to the spaced-apart position.

(86) Turning to FIGS. **15B**, **15B'**, and **16B**, in some embodiments, the distal ends of outer housings **118**, **128** of jaw members **110**, **120** define cut-outs that form shelves **118b**, **128b** between the distal ends of outer housings **118**, **128** and the body portions **118a**, **128a** of housings **118**, **128**, respectively. Shelves **118b**, **128b**, as shown in FIGS. **15B**, **15B'**, and **16B** facilitate the retention of tissue via the distal ends of outer housings **118**, **128**, thus inhibiting slipping of tissue and facilitating spreading and/or dissecting tissue.

(87) Referring to FIGS. **15C**, **15C'**, and **16C**, in some embodiments, in addition to or as an alternative to including shelves **118b**, **128b**, the distal ends of outer housings **118**, **128** of jaw members **110**, **120** define extensions **118c**, **128c** that are relatively narrow and relatively small-radiused as compared to body portions **118a**, **128a** of housings **118**, **128**. These extensions **118c**, **128c** facilitate pressing the distal ends of jaw members **110**, **120** further into tissue (see FIG. **15C**) to ensure a relatively large contact area of tissue against shelves **118b**, **128b** upon moving jaw members **110**, **120** to the spaced-apart position (see FIG. **16C**), thus facilitating the spreading and/or dissecting of tissue.

(88) With reference to FIGS. **15D**, **15D'**, and **16D**, in some embodiments, the distal ends of outer housings **118**, **128** of jaw members **110**, **120** are cut-back to define angled surfaces **118d**, **128d** that define an angle **1''** relative to the perpendicular extending from the distal ends of jaw members **110**, **120** (see FIG. **16D**). Angled surfaces **118d**, **128d**, similarly as with the previous embodiments, facilitate the pressing of the distal ends of jaw members **110**, **120** further into tissue (see FIG. **15D**) as well as the retention of tissue while spreading and/or dissecting tissue.

(89) Referring again to FIGS. **7-12**, trigger assembly **70**, as mentioned above, is operably coupled to knife assembly **170** to enable selective translation of knife blade **174** of knife assembly **170** relative to end effector assembly **100**. Trigger assembly **70** includes trigger **72** and a linkage **76**. Trigger **72** includes a grasping portion **75a** which includes the concave trigger surface **73**, a pivot extension **75b** extending upwardly from grasping portion **75a**, and a proximal extension **75c** extending proximally from grasping portion **75a**. Grasping portion **75a** also includes a tab **75d** extending distally therefrom. Tab **75d** defines an aperture **75e** configured to retain movable end **71b** of biasing member **71** therein. As noted above, fixed end **71a** of biasing member **71** is engaged via retention pin **29f** of first housing component **22a** of housing **20**. In this manner, biasing member **71** serves to bias trigger **72** distally towards an un-actuated position (FIG. **35**).

(90) Pivot extension **75b** of trigger **72** is pivotably coupled to housing **20** via pivot pin **48**, which is engaged within and extends between pivot apertures **29c** of first and second housing components **22a**, **22b** of housing **20**. It is noted that pivot pin **48** is shared by both trigger **72** and movable handle **40**; that is, both trigger **72** and movable handle **40** are pivotable about the same point relative to housing **20**. Proximal extension **75c** of trigger **72** includes a post **75f** that, as detailed below, is operably engaged within cam slot **77e** of linkage **76**.

(91) Linkage **76** serves to operably couple trigger **72** with knife assembly **170** such that pivoting of trigger **72** from the un-actuated position (FIG. **35**) to the actuated position (FIG. **36**) advances knife blade **174** relative to end effector assembly **100** to cut tissue grasped between jaw members **110**, **120**, as detailed below. Linkage **76** defines a generally triangular-shaped configuration including an apex **77a** pointing in a distal direction and a base defining upper and lower corners **77b**, **77c**, respectively. Apex **77a** includes a peg **77d** that is configured for receipt within pivot boss **29e** of first housing component **22a** to pivotably couple linkage **76** relative to housing **20** about apex **77a** thereof. A cam slot **77e**, **77f** is defined through linkage **76** adjacent upper and lower corners **77b**, **77c**, respectively. A coupling pin **78** operably couples cam slot **77e** with knife plate **172** of knife assembly **170**. More specifically, coupling pin **78** includes a cap **79a** defining a slot **79c** (FIG. **9**) configured to receive finger **173** of knife plate **172** and a rod **79b** that is operably engaged within

cam slot **77e**. As noted above, post **75f** of proximal extension **75c** of trigger **72** is operably engaged within cam slot **77f**.

(92) As a result of the above-detailed configuration of trigger assembly **70**, pivoting of trigger **72** between the un-actuated and actuated positions (FIGS. **35** and **36**, respectively) urges linkage **76** to pivot relative to housing **20** ultimately such that coupling pin **78** is urged to translate longitudinally within and relative to housing **20**. As finger **173** of knife plate **172** is engaged with coupling pin **78**, such longitudinal translation of coupling pin **78** is imparted to knife plate **172** for translating knife blade **174** between retracted and extended positions (FIGS. **29-34** and **38-40**, respectively) relative to end effector assembly **100**, as detailed below.

(93) Knife assembly **170**, as noted above, includes a knife plate **172** defining a finger **173** at the proximal end thereof. Knife plate **172** extends distally through housing **20** and shaft **80** atop drive plate **142** and is slidably engaged therewith via receipt of each end of knife plate **172** within track edges **146** of drive plate **142**. Knife assembly **170** further includes knife blade **174** integrally formed with or otherwise engaged to knife plate **172** and extending distally therefrom. Knife blade **174** defines a width less than the combined thickness of jaw members **110**, **120** at the proximal ends thereof but greater than or equal to the combined thickness of jaw members **110**, **120** at the distal ends thereof. Knife blade **174** further defines an elongated opening **176** extending longitudinally therethrough. Elongated opening **176** permits knife blade **174** to be slidably disposed about pivot pin **103** and cam pin **105**. More specifically, elongated opening **176** defines a first portion **177a** having a first width configured to slidably receive pivot pin **103** and a second portion **177b** having a second width configured to slidably receive cam pin **105** but sufficiently small to inhibit receipt of the larger-diameter pivot pin **103** therein.

(94) As appreciated in view of the above, handle assembly **30**, slider assembly **150** of drive assembly **1450**, and trigger assembly **70** enable efficient assembly of instrument **10** in that these components may be operably positioned within housing **20** and relative to one another via a top-down assembly process.

(95) Turning now to FIGS. **22-40**, the use and operation of instrument **10** is described. Initially, as illustrated in FIGS. **22-24**, **29**, **31**, and **33**, movable handle **40** is disposed in the initial position and, correspondingly, jaw members **110**, **120** are disposed in the spaced-apart position. More specifically, with movable handle **40** in the initial position, engagement bulge **51** is disposed in a distal-most position such that slider assembly **150** is disposed in a distal-most position. With slider assembly **150** disposed in its distal-most position, torsion spring **160** is less-tensioned and second leg **162** of torsion spring **160** retains drive plate **142** in a distal-most position. Torsion spring **160** is less-tensioned but not fully un-tensioned in the initial position of movable handle **40**. This configuration maintains a pre-load on drive assembly **140** such that, as noted above, backlash due to the complete removal of tension from torsion spring **160** as jaw members **110**, **120** move from the approximated position back to the spaced-apart position is eliminated. Drive plate **142** is inhibited from moving proximally relative to slider assembly **150** in this position due to the abutment of the proximal edge **145** of drive plate **142** with abutment rib **153d** of proximal housing **152** of slider assembly **150**. Further, in this distal-most position of drive plate **142**, drive plate **142** maintains cam pin **105** at the distal ends of cam slots **114b**, **124b** and, thus, jaw members **110**, **120** are maintained in the spaced-apart position.

(96) Turning for the moment to FIG. **22A**, in some embodiments, rather than first leg **161** of torsion spring **160** (FIGS. **10-11**) being fixed relative to proximal housing **152** of slider assembly **150**, a torsion spring **3160** may be provided including a first leg **3161** that extends through a slot **3153b** defined within proximal housing **152** and is positioned to abut a block **3020** mounted within or monolithically formed with housing **20**. As a result of this configuration, torsion spring **3160** provides the additional function of biasing slider assembly **150** distally, thereby biasing jaw members **110**, **120** towards the spaced-apart position (see FIG. **14**). That is, upon movement of movable handle **40** to translate slider assembly **150** proximally (see FIG. **9**), first leg **3161** of

torsion spring **3160** is maintained in position via its abutment with block **3020**, thereby further tensioning torsion spring **3160** such that, upon release of movable handle **40** (FIG. 9), torsion spring **3160** serves to bias slider assembly **150** distally, thereby biasing jaw members **110**, **120** towards the spaced-apart position (see FIG. 14). Further, multiple blocks **3020** may be provided at different positions within housing **20** such that, during assembly, first leg **3161** of torsion spring **3160** may be positioned to abut a selected one of the blocks **3020** to achieve a desired biasing force.

(97) Returning to FIGS. 22-24, 29, 31, and 33, trigger **72** is initially disposed in the un-actuated position and, accordingly, knife blade **174** is disposed in the retracted position. More specifically, in the un-actuated position, trigger **72** is disposed in a distal-most position under the bias of biasing member **71** such that lower corner **77c** of linkage **76**, which is coupled to trigger **72** via engagement of post **75f** within slot **77f**, is disposed in a distal-most position. Since upper and lower corners **77b**, **77c** of linkage **76** are disposed on opposite sides of apex **77a**, with lower corner **77c** disposed in its distal-most position, upper corner **77b** is disposed in a proximal-most position. With upper corner **77b** disposed in its proximal-most position, knife plate **172** is likewise disposed in a proximal-most position due to the engagement of pin **78** within slot **77e**. The proximal-most position of knife plate **172** corresponds to the retracted position of knife blade **174**, wherein knife blade **174** is disposed between flanges **113a**, **123a** of jaw members **110**, **120** but does not extend distally therefrom so as to avoid interference with tissue disposed between jaw members **110**, **120**. Further, in this position, pivot pin **103** is disposed at the distal end of first portion **177a** of opening **176** of knife blade **174** and cam pin **105** is likewise disposed within first portion **177a** of opening **176**.

(98) With additional reference to FIGS. 25-28, 30, 32, and 34, in order to move jaw members **110**, **120** to the approximated position to grasp tissue therebetween, movable handle **40** is pulled proximally towards fixed handle portion **26** from the initial position (FIG. 22) to the compressed position (FIG. 27). Upon such movement of movable handle **40** to the compressed position, engagement bulge **51** of movable handle **40** moves proximally relative to housing **20**, thereby urging slider assembly **150** proximally through housing **20**. Torsion spring **160**, in the less-tensioned state, is translated proximally together with slider assembly **150** such that second leg **162** of torsion spring **160** pulls drive plate **142** proximally in connection with the proximal translation of slider assembly **150**. In other words, at this point, slider assembly **150** and drive plate **142** move in concert with one another. As drive plate **142** is pulled proximally, cam pin **105** is pulled proximally through cam slots **114b**, **124b** such that jaw members **110**, **120** are pivoted from the spaced-apart position to the approximated position to grasp tissue therebetween.

(99) As detailed above, movement of movable handle **40** from the initial position (FIG. 22) to the compressed position (FIG. 27) similarly translates drive plate **142** proximally, thereby moving jaw members **110**, **120** to the approximated position to grasp tissue therebetween. Drive plate **142** is still inhibited from moving proximally relative to slider assembly **150** in this position due to the abutment of the proximal edge **145** of drive plate **142** with abutment rib **153d** of proximal housing **152** of slider assembly **150**.

(100) At this point, with tissue grasped between jaw members **110**, **120**, instrument **10** may be utilized as a pliers to maneuver, manipulate, and/or reposition tissue. In particular, as noted above and illustrated in FIGS. 15 and 16, jaw members **110**, **120** may be approximated adjacent a wall of tissue to pinch tissue between the distal tips of jaw members **110**, **120** to enable maneuvering, manipulating, and/or repositioning thereof.

(101) Referring to FIG. 35, in order to apply energy to tissue grasped between jaw members **110**, **120** to treat tissue, movable handle **40** is compressed further towards fixed handle portion **26** to an activation position, wherein an appropriate closure force or closure force within an appropriate range, is achieved and energy activation is initiated. As movable handle **40** is moved further proximally relative to housing **20** beyond the compressed position, an appropriate closure force or closure force within an appropriate range is imparted to tissue grasped between electrically-

conductive plates **112**, **122** of jaw members **110**, **120** regardless of the thickness or compressibility of tissue or the position of movable handle **40**. This is because, upon movement of movable handle **40** from the compressed position towards the activation position, slider assembly **150** is translated proximally while drive plate **142** is maintained in position. In other words, upon movement of movable handle **40** from the compressed position to the activated position, slider assembly **150** and drive plate **142** no longer move in concert with one another. Rather, as detailed below, slider assembly **150** and drive plate **142** are decoupled to permit relative motion therebetween.

(102) The decoupling of slider assembly **150** and drive plate **142** to permit relative motion therebetween is provided via torsion spring **160**. More specifically, upon proximal movement of movable handle **40**, a first force is imparted from movable handle **40**, through slider assembly **150**, first leg **161** of torsion spring **160**, the body of torsion spring **160**, and second leg **162** of torsion spring **160**, to drive plate **142** to urge drive plate **142** in a proximal direction, while a second, opposite force acts on drive plate **142** and, thus, second leg **162** of torsion spring **160** in a distal direction to resist further compression of tissue between jaw members **110**, **120**. Once the second, opposite force exceeds the spring force of torsion spring **160**, proximal movement of slider assembly **150** no longer results in proximal movement of drive plate **142** but, rather, results in further tensioning of torsion spring **160**, which absorbs the force imparted thereto from movement of movable handle **40**. Thus, once this point has been reached, further proximal translation of slider assembly **150** urges first end **161** of torsion spring **160** proximally, while second opposite force retains second leg **162** of torsion spring **160** in position, thereby further tensioning torsion spring **160**. Since second leg **162** of torsion spring **160** is retained in position, drive plate **142** is likewise retained in position despite the proximal translation of movable handle **40** and slider assembly **150**.

(103) It is noted that, during movement of movable handle **40** from the initial position to the compressed position, as detailed above, the second, opposite force is less than the spring force of torsion spring **160** and, thus, slider assembly **150**, first and second legs **161**, **162** of torsion spring **160**, and drive plate **142** move in conjunction with one another. Further, torsion spring **160** may be configured such that the second, opposite force exceeds the spring force of torsion spring **160** at a force corresponding to a closure pressure on tissue between 3 kg/cm^{sup.2} to 16 kg/cm^{sup.2}, as it has been found that closure forces within this range facilitate sealing of tissue grasped between jaw members **110**, **120**. However, other forces and/or force ranges are also contemplated, e.g., for treating tissue in other manners (coagulating, cauterizing, etc.).

(104) Continuing with reference to FIG. 35, upon achieving the activation position of movable handle **40**, button activation post **196** (FIG. 7) of movable handle **40** contacts depressible button **192** sufficiently so as to depress depressible button **192** into fixed handle portion **26** to activate switch **194**. Switch **194**, as noted above, is disposed in electrical communication with the generator (not shown) and electrically-conductive plates **112**, **122** of jaw members **110**, **120**, respectively, such that activation of switch **194** initiates the supply of energy to electrically-conductive plates **112**, **122** to treat, e.g., coagulate, cauterize, and/or seal, tissue grasped therebetween.

(105) Referring to FIGS. 36-40, once tissue has been treated or where it is only desired to cut tissue, knife blade **174** may be advanced between jaw members **110**, **120** to cut tissue grasped therebetween. In order to advance knife blade **174** from the retracted position to the extended position, trigger **72** is pulled proximally against the bias of biasing member **71** from the un-actuated position to the actuated position. As trigger **72** is pulled proximally, post **75f** is pushed proximally to urge linkage **76** to pivot counter-clockwise (from the orientation illustrated in FIG. 36) such that upper corner **77b** of linkage **76** is moved distally. Distal movement of upper corner **77b** urges pin **78** to translate distally due to the engagement of rod **79b** of pin **78** within slot **77e** of upper corner **77b** of linkage **76**. Distal translation of pin **78**, in turn, urges knife plate **172** distally due to the engagement of finger **173** of knife plate **172** within cap **79a** of pin **78**.

(106) As detailed above, movement of trigger **72** from the un-actuated position to the actuated position urges knife plate **172** distally. More specifically, knife plate **172** is urged distally such that

knife blade **174** is advanced distally from the retracted position to the extended position. As knife blade **174** is advanced distally, knife blade **174** enters the knife slots of jaw members **110**, **120** defined by the respective knife slots **112b**, **122b** of electrically-conductive plates **112**, **122** and the knife slots **116c** of the respective spacers **115** (only spacer **115** of jaw member **110** is illustrated). As can be appreciated, translation of knife blade **174** through the knife slots of jaw members **110**, **120** to the extended position thereof divides tissue grasped between jaw members **110**, **120**.

(107) Due to the fact that knife blade **174** defines a width greater than or equal to the combined thickness of jaw members **110**, **120** at the distal ends thereof, as knife blade **174** is advanced distally through the knife slots, knife blade **174** may extend at least partially through windows **119**, **129** and openings **131**, **132** of jaw members **110**, **120**, depending upon the thickness of tissue grasped between jaw members **110**, **120** (see FIG. 39). Further, as knife blade **174** is advanced distally, pivot pin **103** and cam pin **105** translate proximally along opening **176** eventually such that cam pin **105** extends through second portion **177b** of opening **176**. As pivot pin **103** is too large to extend into second portion **177b** of opening **176**, interference therebetween defines the distal-most extent of travel of knife blade **174**. However, other components of knife assembly **170** and/or trigger assembly **70** may additionally or alternatively inhibit the extension of knife blade **174**.

(108) Upon release of trigger **72**, trigger **72** and knife plate **172** are returned proximally under the bias of biasing member **71** such that knife blade **174** is returned to the retracted position.

Thereafter, movable handle **40** may be returned to the initial position to release the treated and/or divided tissue.

(109) The various embodiments disclosed herein may also be configured to work with robotic surgical systems and what is commonly referred to as “Telesurgery.” Such systems employ various robotic elements to assist the surgeon and allow remote operation (or partial remote operation) of surgical instrumentation. Various robotic arms, gears, cams, pulleys, electric and mechanical motors, etc. may be employed for this purpose and may be designed with a robotic surgical system to assist the surgeon during the course of an operation or treatment. Such robotic systems may include remotely steerable systems, automatically flexible surgical systems, remotely flexible surgical systems, remotely articulating surgical systems, wireless surgical systems, modular or selectively configurable remotely operated surgical systems, etc.

(110) The robotic surgical systems may be employed with one or more consoles that are next to the operating theater or located in a remote location. In this instance, one team of surgeons or nurses may prep the patient for surgery and configure the robotic surgical system with one or more of the instruments disclosed herein while another surgeon (or group of surgeons) remotely control the instruments via the robotic surgical system. As can be appreciated, a highly skilled surgeon may perform multiple operations in multiple locations without leaving his/her remote console which can be both economically advantageous and a benefit to the patient or a series of patients.

(111) The robotic arms of the surgical system are typically coupled to a pair of master handles by a controller. The handles can be moved by the surgeon to produce a corresponding movement of the working ends of any type of surgical instrument (e.g., end effectors, graspers, knives, scissors, etc.) which may complement the use of one or more of the embodiments described herein. The movement of the master handles may be scaled so that the working ends have a corresponding movement that is different, smaller or larger, than the movement performed by the operating hands of the surgeon. The scale factor or gearing ratio may be adjustable so that the operator can control the resolution of the working ends of the surgical instrument(s).

(112) The master handles may include various sensors to provide feedback to the surgeon relating to various tissue parameters or conditions, e.g., tissue resistance due to manipulation, cutting or otherwise treating, pressure by the instrument onto the tissue, tissue temperature, tissue impedance, etc. As can be appreciated, such sensors provide the surgeon with enhanced tactile feedback simulating actual operating conditions. The master handles may also include a variety of different actuators for delicate tissue manipulation or treatment further enhancing the surgeon's ability to

mimic actual operating conditions.

(113) From the foregoing and with reference to the various figure drawings, those skilled in the art will appreciate that certain modifications can also be made to the present disclosure without departing from the scope of the same. While several embodiments of the disclosure have been shown in the drawings, it is not intended that the disclosure be limited thereto, as it is intended that the disclosure be as broad in scope as the art will allow and that the specification be read likewise. Therefore, the above description should not be construed as limiting, but merely as exemplifications of particular embodiments. Those skilled in the art will envision other modifications within the scope and spirit of the claims appended hereto.

Claims

1. An electrosurgical end effector assembly, comprising: first and second jaw members configured to grasp tissue therebetween, the first jaw member including: a jaw frame defining a proximal portion and a distal portion, the proximal portion of the jaw frame including first and second spaced-apart flanges defining aligned pivot apertures configured to receive a pivot pin to pivotably couple the first and second jaw members with one another and aligned cam slots configured to receive a cam pin to drive pivoting of the first jaw member relative to the second jaw member, the distal portion of the jaw frame defining an upright and a base, wherein the upright and the base are substantially perpendicular relative to one another; an electrode configured to conduct energy to tissue, the electrode including a leg extending in substantially parallel orientation relative to the upright, the leg defining a plurality of apertures spaced-apart along at least a portion of a length of the leg; and an insulative housing extending along the leg and through the plurality of apertures of the leg to secure the electrode relative to the insulative housing.
2. The electrosurgical end effector assembly according to claim 1, wherein the first jaw member defines a tunnel extending longitudinally along a portion of the leg and wherein a lead wire extends through the tunnel and electrically connects with the leg.
3. The electrosurgical end effector assembly according to claim 1, further comprising a clevis operably receiving the first and second spaced-apart flanges of the proximal portion of the jaw frame.
4. The electrosurgical end effector assembly according to claim 1, wherein the first and second jaw members are configured to conduct electrosurgical energy through tissue grasped therebetween.
5. The electrosurgical end effector assembly according to claim 1, wherein the electrode of the first jaw member and an electrode of the second jaw member are configured to conduct electrosurgical energy through tissue grasped therebetween.
6. The electrosurgical end effector assembly according to claim 1, wherein the electrode includes an exposed portion that is exposed above the insulative housing.
7. The electrosurgical end effector assembly according to claim 6, wherein the exposed portion of the electrode defines a tissue contacting surface.
8. The electrosurgical end effector assembly according to claim 1, wherein the jaw frame is stainless steel.
9. The electrosurgical end effector assembly according to claim 1, wherein the insulative housing is a first overmold.
10. The electrosurgical end effector assembly according to claim 9, wherein the first jaw member further includes a second overmold.
11. An electrosurgical end effector assembly, comprising: first and second jaw members configured to grasp tissue therebetween, the first jaw member including: a jaw frame defining a proximal portion and a distal portion, the proximal portion of the jaw frame including first and second spaced-apart flanges defining aligned pivot apertures configured to receive a pivot pin to pivotably couple the first and second jaw members with one another and aligned cam slots configured to

receive a cam pin to drive pivoting of the first jaw member relative to the second jaw member, the distal portion of the jaw frame defining an upright and a base, wherein the upright and the base are substantially perpendicular relative to one another; an electrode configured to conduct energy to tissue, the electrode including a leg defining an upper end and a lower end, the leg extending in substantially parallel orientation relative to the upright, the leg defining a plurality of apertures spaced-apart along at least a portion of a length of the leg and disposed between the upper and lower ends of the leg, wherein the upper end of the leg is exposed; and an insulative spacer capturing the lower end of the leg within the insulative spacer, the insulative spacer disposed on the jaw frame and positioned to electrically isolate the electrode from the jaw frame.

12. The electrosurgical end effector assembly according to claim 11, wherein the insulative spacer defines a tunnel extending longitudinally along a portion of the leg and wherein a lead wire extends through the tunnel and electrically connects with the leg.

13. The electrosurgical end effector assembly according to claim 11, wherein the first and second jaw members are configured to conduct electrosurgical energy through tissue grasped therebetween.

14. The electrosurgical end effector assembly according to claim 11, wherein the electrode of the first jaw member and an electrode of the second jaw member are configured to conduct electrosurgical energy through tissue grasped therebetween.

15. The electrosurgical end effector assembly according to claim 11, wherein the electrode includes an exposed portion that is exposed above the insulative spacer.

16. The electrosurgical end effector assembly according to claim 11, wherein the upright and the base of the jaw frame define an interior angle side of the jaw frame and an exterior angle side of the jaw frame, and wherein the insulative spacer is at least partially disposed on the interior angle side of the jaw frame.

17. The electrosurgical end effector assembly according to claim 16, further comprising an insulative housing disposed on the exterior angle side of the jaw frame.

18. The electrosurgical end effector assembly according to claim 11, wherein the insulative spacer is a first overmold.

19. The electrosurgical end effector assembly according to claim 18, wherein the first jaw member further includes a second overmold.

20. The electrosurgical end effector assembly according to claim 19, wherein the second overmold is an insulative housing.
